

Assembly Guide for the Hartmann Sensor

1 Assembling the Hartmann Sensor

Before handling any optical or electronic components, put on a pair of clean gloves.

Important

Always wear gloves during the assembly process to prevent fingerprints and dust on the optical and electronic components.

1.1 Step 1. Install the Camera PCB

Take the front housing, which can be identified by the groove for the camera PCB and the SM1 opening.

Place the camera PCB into the groove. Make sure the sensor faces the SM1 opening.

The PCB may move slightly at this stage. This is normal and it will be secured after the back housing is installed.

1.2 Step 2. Install the Back Housing

Place the back housing onto the camera PCB.

Align the opening in the back housing with the metal connector on the camera PCB.

Insert the four screws and tighten each screw a few turns. After all four screws are in place, tighten them gradually.

Important

The housing may slightly press on the camera chip. When you feel resistance, stop tightening. Do not tighten the screws with strong force.

Starting Point for This Course

Starting Point

In this course, your Hartmann sensor housing may already be assembled.

If you receive a sensor body like the one shown below, you do not need to repeat Step 1 and Step 2.

You can start directly from **Step 3. Assemble the Lens Tube**.

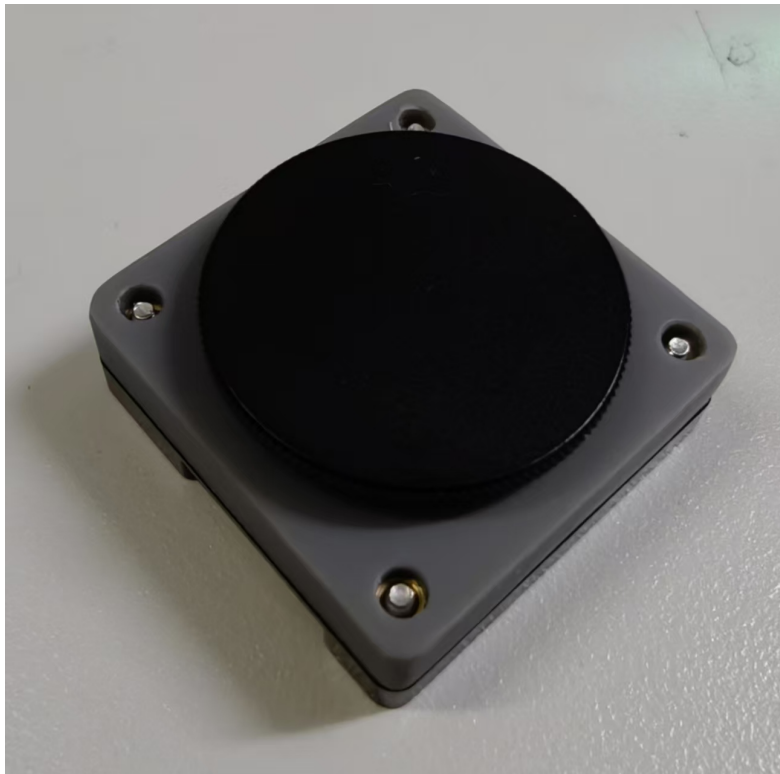


Figure 1: Assembled Hartmann sensor housing without the lens tube. If you receive this part, you can start from the lens tube assembly step.

1.3 Step 3. Assemble the Lens Tube

Use the SM1 spanner wrench to install one retaining ring into the lens tube.

Screw it near the bottom of the lens tube. It does not need to be tight.

Place the mounted MLA into the lens tube. The MLA has no front or back side, so either side can face the camera.

Install the second retaining ring. Tighten it until the MLA no longer moves.

Important

Do not tighten the retaining ring too much. Stop when you feel a little resistance.

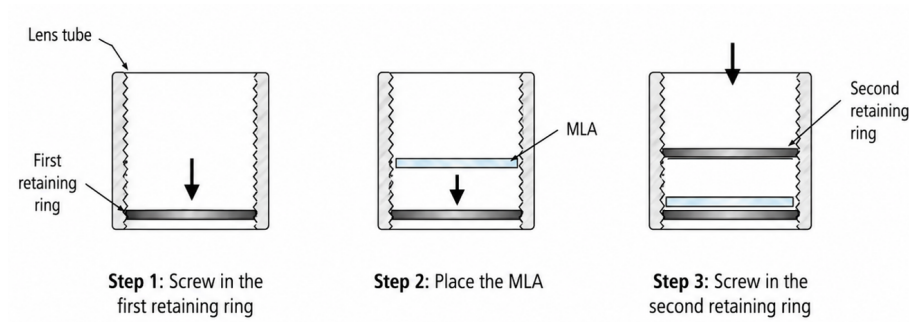


Figure 2: Assembly of the MLA inside the lens tube.

1.4 Step 4. Attach the Lens Tube

Screw the assembled lens tube into the SM1 opening on the front housing.

Turn the lens tube slowly. If the thread feels tight because of the 3D printed housing, you can stop when you feel clear resistance.

The lens tube does not need to be fully tightened or screwed all the way to the end.

Important

The housing is 3D printed, so the thread may feel tight. Do not force the lens tube. Stop when you feel clear resistance. The lens tube does not need to be screwed all the way in.

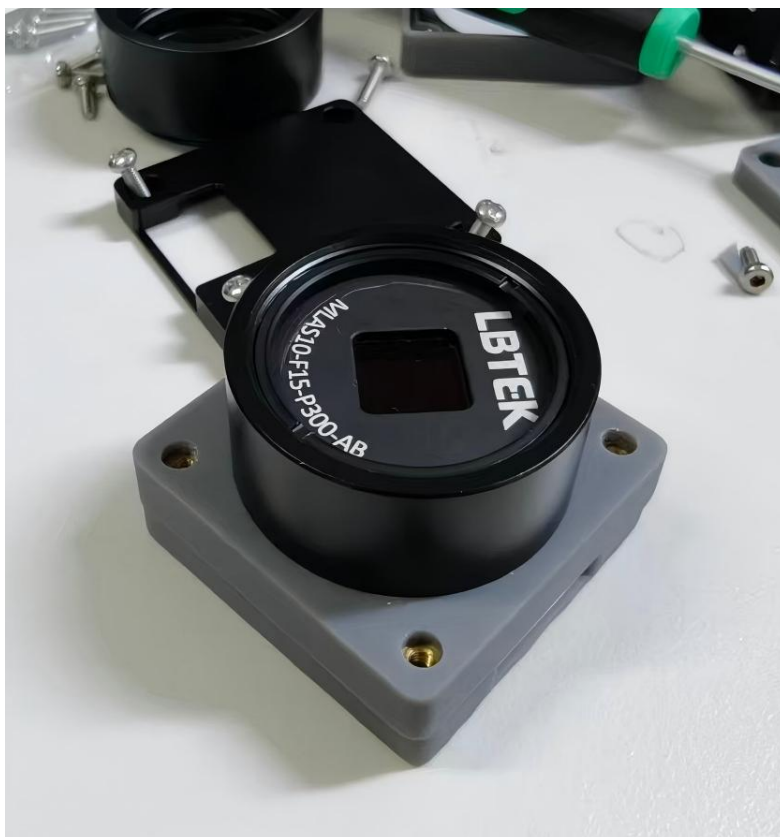


Figure 3: Attach the assembled lens tube to the front housing.

1.5 Step 5. Connect the Camera

Connect the camera to your computer and open the **Galaxy Viewer** software.

The software icon looks like this:



Figure 4: Galaxy Viewer software icon.

After opening the software, find the camera in the left device list.

Double-click the camera name to open it.

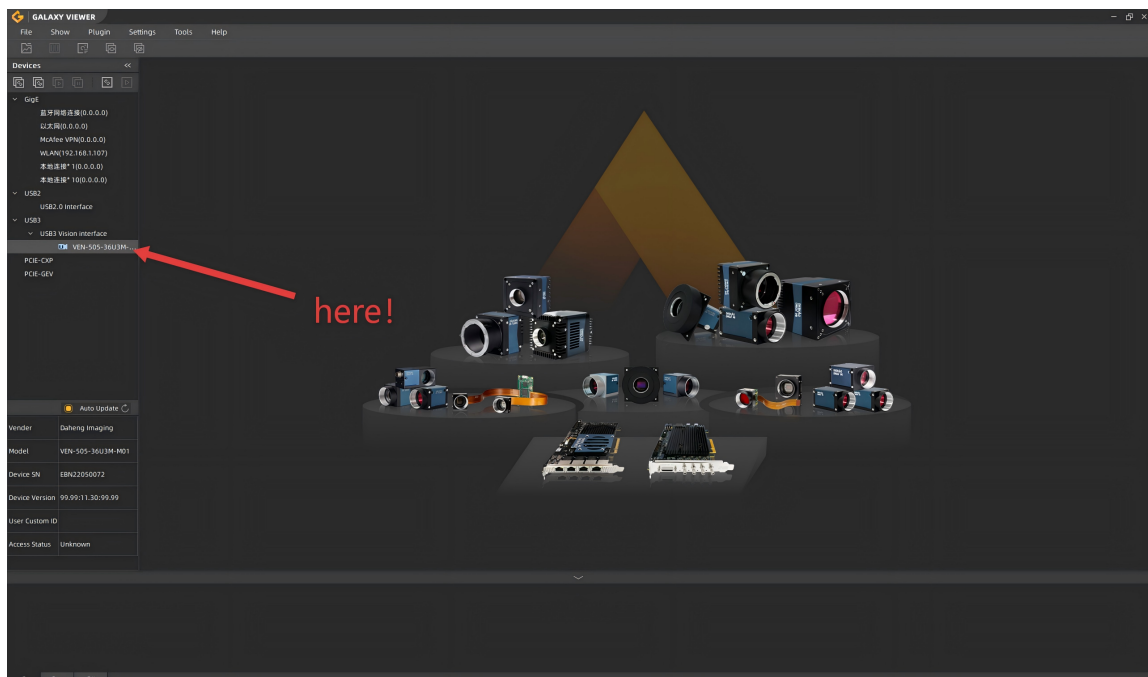


Figure 5: Find the camera in the device list and double-click it.

After the camera window opens, click the acquisition button on the top toolbar to start imaging.

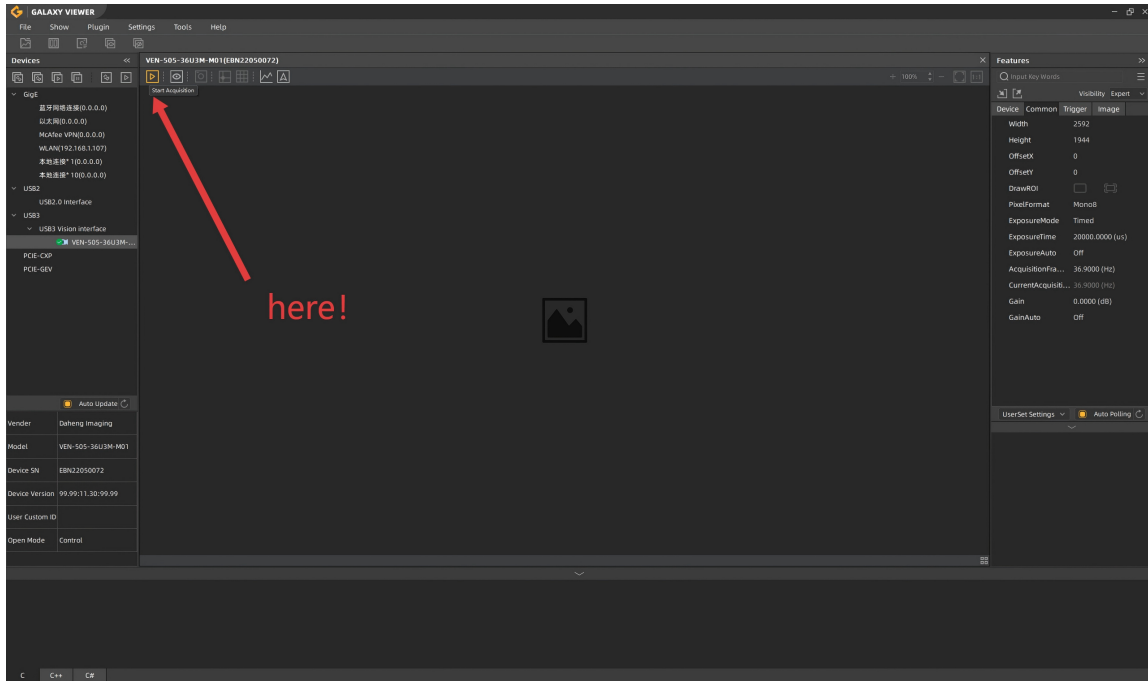


Figure 6: Click the acquisition button to start image capture.

If the camera is working correctly, you should see the live image in the main window.

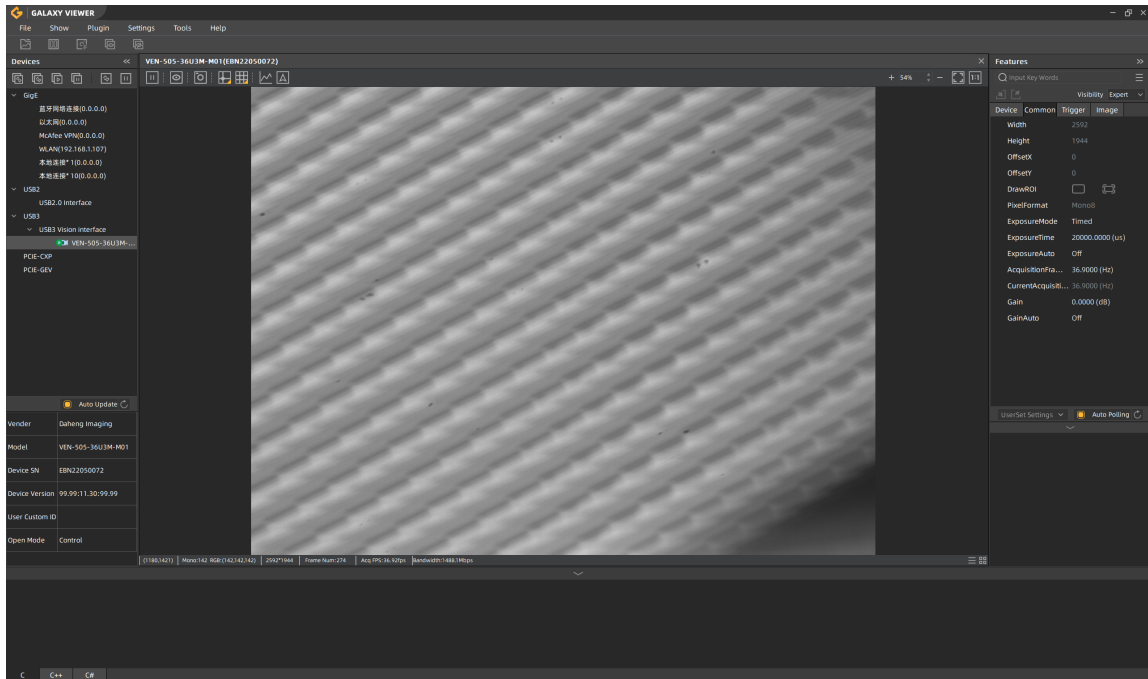


Figure 7: Live camera image shown in Galaxy Viewer.

Note

Galaxy Viewer is available for Windows and Linux only. If you are using macOS, a Linux virtual machine is required. A Windows computer with Galaxy Viewer installed will be provided during the course.

1.6 Step 6. Check the Spot Pattern

Place the Hartmann sensor in the prepared optical setup with a collimated beam.

Adjust the position of the sensor until the beam passes through the center of the MLA.

You should see a regular array of large bright spots on the screen.

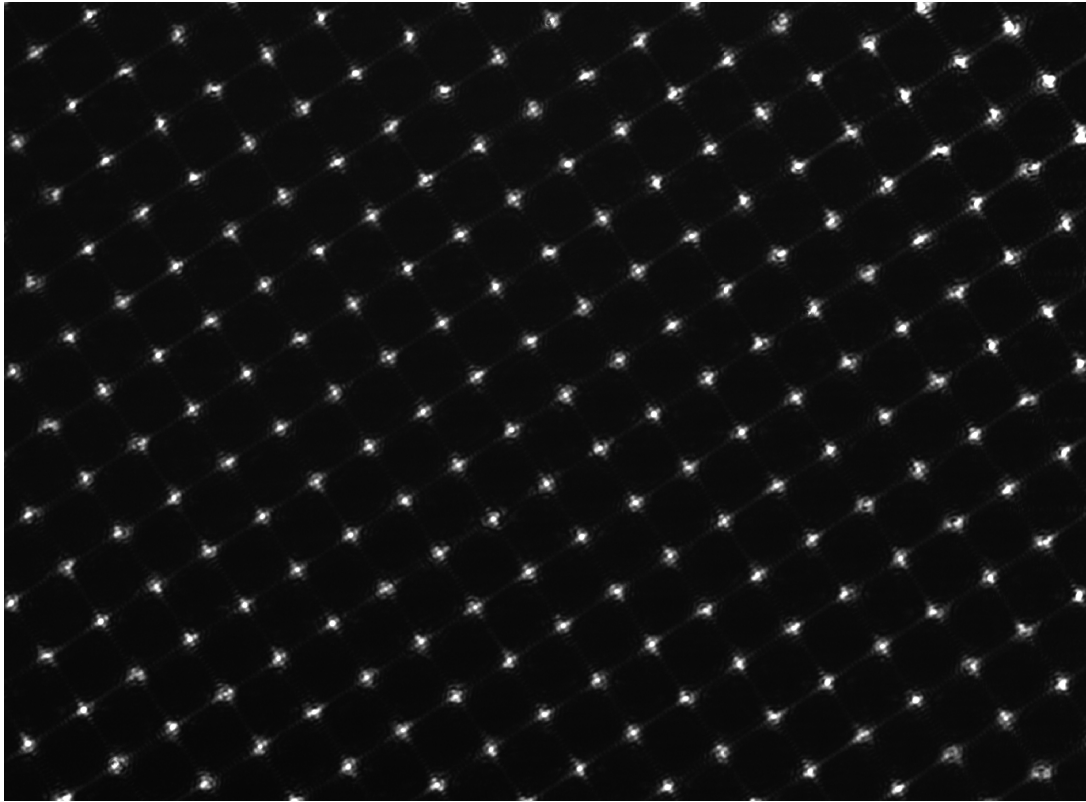


Figure 8: Initial spot pattern after installing the MLA. At this stage, the MLA has not been fully adjusted yet, so the spot array may be tilted and the spots may not be at the best focus.

1.7 Step 7. Focus Adjustment

Unscrew the lens tube from the housing.

Remove the outer retaining ring and take out the MLA. Move the inner retaining ring by about two turns, then put the MLA back and install the outer retaining ring again.

Screw the lens tube back onto the housing and check the spot pattern in the Galaxy software.

Repeat these steps until the spots become as small as possible.

1.8 Step 8. Align the MLA

Keep the inner retaining ring in place.

Loosen the outer retaining ring slightly and rotate the MLA until its edges are parallel to the edges of the square housing.

Tighten the outer retaining ring again. It only needs to hold the MLA in place, so do not tighten it too much.

2 Final Expected Result

After the Hartmann sensor is assembled and adjusted correctly, you should see a regular spot array like the image below.

The spots should be small, clear, and arranged in a regular grid. This means the MLA and the camera are well aligned.

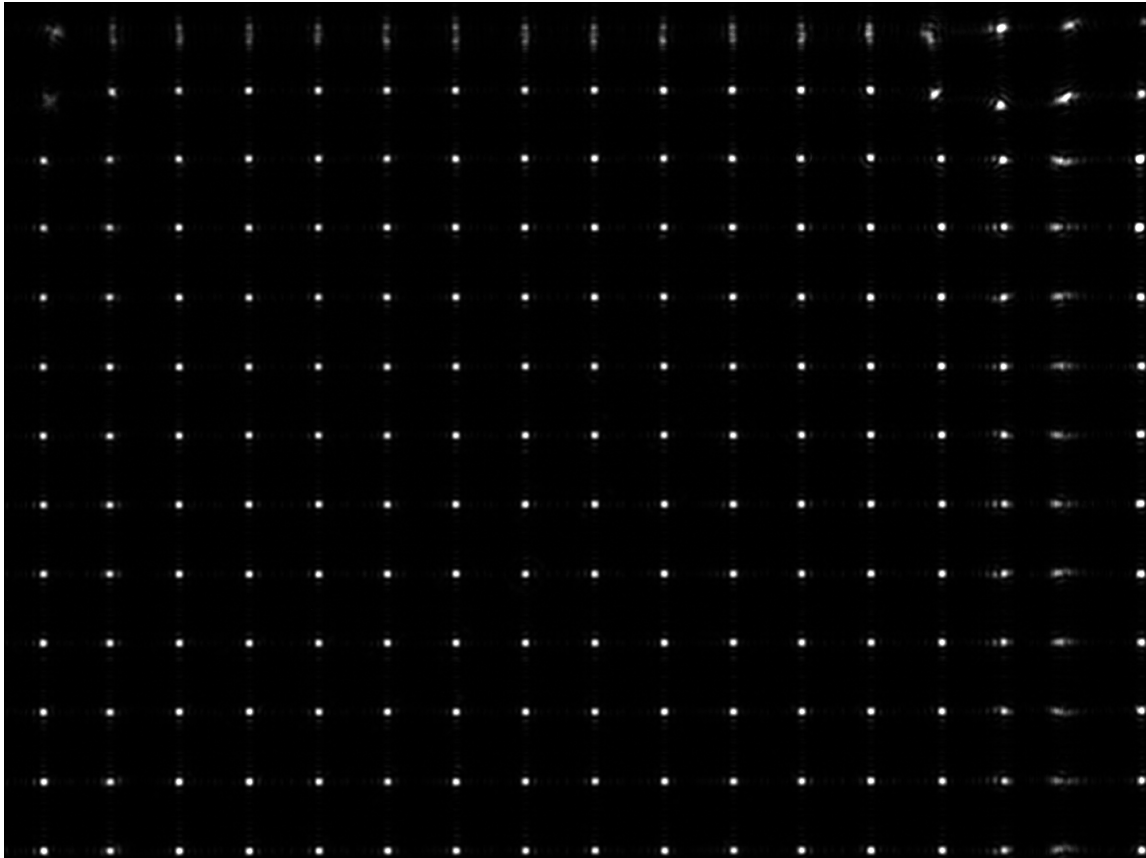


Figure 9: Final expected Shack-Hartmann spot pattern after assembly and adjustment.

Have Fun!

You have now finished the Hartmann sensor assembly and the basic Shack-Hartmann analysis tutorial.

We hope this tutorial helps you understand how a Hartmann sensor works and how to assemble it.

Have fun with your Hartmann sensor!